

Python Syntax

Python is known for its clear and readable syntax. Here are some key aspects of Python syntax:

Indentation:

- Python uses indentation to define blocks of code.
- Consistent use of indentation is crucial as it indicates the scope of loops, functions, and other constructs.

Example:

Correct:

```
if True:
```

```
    print("This is indented")
```

```
if 5 > 2:
```

```
    print("So is this")
```

Incorrect:

```
if True:
```

```
print("This is not indented properly")
```

```
if 5 > 2:
```

```
    print("This will cause an error")
```

Python Comments

Comments can be used to explain Python code.

Comments can be used to make the code more readable.

Comments can be used to prevent execution when testing code.

Types of Comment Lines

1. Single Line Comment (#)

Example:

```
# This is a single-line comment
```

2. Multi Line Comment (Triple quotes - """)

Example:

```
"""
```

```
This is a
```

```
multi-line comment
```

```
"""
```

Python Variables

Variables are containers for storing data values.

Variable Names

A variable can have a short name (like x and y) or a more descriptive name (age, carname, total_volume). Rules for Python variables:

- A variable name must start with a letter or the underscore character
- A variable name cannot start with a number
- A variable name can only contain alpha-numeric characters and underscores (A-z, 0-9, and _)
- Variable names are case-sensitive (age, Age and AGE are three different variables)
- A variable name cannot be any of the Python keywords.

Example

Legal variable names:

```
myvar = "John"  
my_var = "John"  
_my_var = "John"  
myVar = "John"  
MYVAR = "John"  
myvar2 = "John"
```

Example

Illegal variable names:

```
2myvar = "John"  
my-var = "John"  
my var = "John"
```

Multi Words Variable Names

Variable names with more than one word can be difficult to read.

There are several techniques you can use to make them more readable:

Camel Case

Each word, except the first, starts with a capital letter:

```
myVariableName = "John"
```

Pascal Case

Each word starts with a capital letter:

```
MyVariableName = "John"
```

Snake Case

Each word is separated by an underscore character:

```
my_variable_name = "John"
```

Creating Variables

Python has no command for declaring a variable.

A variable is created the moment you first assign a value to it.

Example

```
x = 5  
y = "John"  
print(x)  
print(y)
```

Variables do not need to be declared with any particular *type*, and can even change type after they have been set.

Example

```
x = 4          # x is of type int
x = "Sally"    # x is now of type str
print(x)
```

Get the Type

You can get the data type of a variable with the `type ()` function.

Example

```
x = 5
y = "John"
print(type(x))
print(type(y))
```

Single or Double Quotes?

String variables can be declared either by using single or double quotes:

Example

```
x = "John"
# is the same as
x = 'John'
```

Case-Sensitive

Variable names are case-sensitive.

Example

```
a = 4
A = "Sally"
#A will not overwrite a
```

This will create two variables

Many Values to Multiple Variables

Python allows you to assign values to multiple variables in one line:

Example

```
x, y, z = "Orange", "Banana", "Cherry"  
print(x)  
print(y)  
print(z)
```

Note: Make sure the number of variables matches the number of values, or else you will get an error.

One Value to Multiple Variables

And you can assign the *same* value to multiple variables in one line:

Example

```
x = y = z = "Orange"  
print(x)  
print(y)  
print(z)
```

Python Data Types

Built-in Data Types

In programming, data type is an important concept.

Variables can store data of different types, and different types can do different things.

Python has the following data types built-in by default, in these categories:

Text Type: `str`

Numeric Types: `int, float, complex`

Sequence Types: `list, tuple, range`

Mapping Type: `dict`

Set Types: `set, frozenset`

Boolean Type: `bool`

Binary Types: `bytes, bytearray, memoryview`

None Type: `NoneType`

Setting the Data Type

In Python, the data type is set when you assign a value to a variable:

Example	Data Type
<code>x = "Hello World"</code>	str
<code>x = 20</code>	int
<code>x = 20.5</code>	float
<code>x = 1j</code>	complex
<code>x = ["apple", "banana", "cherry"]</code>	list
<code>x = ("apple", "banana", "cherry")</code>	tuple
<code>x = range(6)</code>	range
<code>x = {"name" : "John", "age" : 36}</code>	dict
<code>x = {"apple", "banana", "cherry"}</code>	set
<code>x = frozenset({"apple", "banana", "cherry"})</code>	frozenset
<code>x = True</code>	bool
<code>x = b"Hello"</code>	bytes

<code>x = bytearray(5)</code>	<code>bytearray</code>
<code>x = memoryview(bytes(5))</code>	<code>memoryview</code>
<code>x = None</code>	<code>NoneType</code>

Setting the Specific Data Type

If you want to specify the data type, you can use the following constructor functions:

Example	Data Type
<code>x = str("Hello World")</code>	<code>str</code>
<code>x = int(20)</code>	<code>int</code>
<code>x = float(20.5)</code>	<code>float</code>
<code>x = complex(1j)</code>	<code>complex</code>
<code>x = list(("apple", "banana", "cherry"))</code>	<code>list</code>

<code>x = tuple(("apple", "banana", "cherry"))</code>	tuple
<code>x = range(6)</code>	range
<code>x = dict(name="John", age=36)</code>	dict
<code>x = set(("apple", "banana", "cherry"))</code>	set
<code>x = frozenset(("apple", "banana", "cherry"))</code>	frozenset
<code>x = bool(5)</code>	bool
<code>x = bytes(5)</code>	bytes

Python Casting

Specify a Variable Type

There may be times when you want to specify a type on to a variable. This can be done with casting. Python is an object-orientated language, and as such it uses classes to define data types, including its primitive types.

Casting in python is therefore done using constructor functions:

- `int()` - constructs an integer number from an integer literal, a float literal (by removing all decimals), or a string literal (providing the string represents a whole number)
- `float()` - constructs a float number from an integer literal, a float literal or a string literal (providing the string represents a float or an integer)
- `str()` - constructs a string from a wide variety of data types, including strings, integer literals and float literals

Example

Integers:

```
x = int(1)    # x will be 1
y = int(2.8)  # y will be 2
z = int("3")  # z will be 3
```

Example

Floats:

```
x = float(1)    # x will be 1.0
y = float(2.8)  # y will be 2.8
z = float("3")  # z will be 3.0
w = float("4.2") # w will be 4.2
```

Example

Strings:

```
x = str("s1")  # x will be 's1'
y = str(2)     # y will be '2'
z = str(3.0)   # z will be '3.0'
```

Python Operators

Operators are used to perform operations on variables and values.

In the example below, we use the `+` operator to add together two values:

Example

```
print(10 + 5)
```

Python divides the operators in the following groups:

- Arithmetic operators
- Assignment operators
- Comparison operators
- Logical operators
- Identity operators
- Membership operators
- Bitwise operators

Python Arithmetic Operators

Arithmetic operators are used with numeric values to perform common mathematical operations:

Operator	Name	Example
+	Addition	$x + y$
-	Subtraction	$x - y$
*	Multiplication	$x * y$
/	Division	x / y

%	Modulus	x % y
**	Exponentiation	x ** y
//	Floor division	x // y

Python Assignment Operators

Assignment operators are used to assign values to variables:

Operator	Example	Same As
=	x = 5	x = 5
+=	x += 3	x = x + 3
-=	x -= 3	x = x - 3
*=	x *= 3	x = x * 3
/=	x /= 3	x = x / 3
%=	x %= 3	x = x % 3

<code>//=</code>	<code>x //= 3</code>	<code>x = x // 3</code>
<code>**=</code>	<code>x **= 3</code>	<code>x = x ** 3</code>
<code>&=</code>	<code>x &= 3</code>	<code>x = x & 3</code>
<code> =</code>	<code>x = 3</code>	<code>x = x 3</code>
<code>^=</code>	<code>x ^= 3</code>	<code>x = x ^ 3</code>
<code>>>=</code>	<code>x >>= 3</code>	<code>x = x >> 3</code>
<code><<=</code>	<code>x <<= 3</code>	<code>x = x << 3</code>
<code>:=</code>	<code>print(x := 3)</code>	<code>x = 3</code> <code>print(x)</code>

Python Comparison Operators

Comparison operators are used to compare two values:

Operator	Name	Example
<code>==</code>	Equal	<code>x == y</code>

!=	Not equal	x != y
>	Greater than	x > y
<	Less than	x < y
>=	Greater than or equal to	x >= y
<=	Less than or equal to	x <= y

Python Logical Operators

Logical operators are used to combine conditional statements:

Operator	Description	Example
and	Returns True if both statements are true	x < 5 and x < 10
or	Returns True if one of the statements is true	x < 5 or x < 4
not	Reverse the result, returns False if the result is true	not(x < 5 and x < 10)

Python Identity Operators

Identity operators are used to compare the objects, not if they are equal, but if they are actually the same object, with the same memory location:

Operator	Description	Example
is	Returns True if both variables are the same object	x is y
is not	Returns True if both variables are not the same object	x is not y

Python Membership Operators

Membership operators are used to test if a sequence is presented in an object:

Operator	Description	Example
in	Returns True if a sequence with the specified value is present in the object	x in y
not in	Returns True if a sequence with the specified value is not present in the object	x not in y

Python Bitwise Operators

Bitwise operators are used to compare (binary) numbers:

Operator	Name	Description	Example
&	AND	Sets each bit to 1 if both bits are 1	x & y
	OR	Sets each bit to 1 if one of two bits is 1	x y
^	XOR	Sets each bit to 1 if only one of two bits is 1	x ^ y
~	NOT	Inverts all the bits	~x
<<	Zero fill left shift	Shift left by pushing zeros in from the right and let the leftmost bits fall off	x << 2
>>	Signed right shift	Shift right by pushing copies of the leftmost bit in from the left, and let the rightmost bits fall off	x >> 2

Operator Precedence

Operator precedence describes the order in which operations are performed.

The precedence order is described in the table below, starting with the highest precedence at the top:

Operator	Description
()	Parentheses
**	Exponentiation
+X -X ~X	Unary plus, unary minus, and bitwise NOT
* / // %	Multiplication, division, floor division, and modulus
+ -	Addition and subtraction
<< >>	Bitwise left and right shifts
&	Bitwise AND
^	Bitwise XOR

	Bitwise OR
== != > >= < <= is is not in not in	Comparisons, identity, and membership operators
not	Logical NOT
and	AND
or	OR

If two operators have the same precedence, the expression is evaluated from left to right.

Example

Multiplication `*` has higher precedence than addition `+`, and therefore multiplications are evaluated before additions:

```
print(100 + 5 * 3)
```

Python Lists

Lists are used to store multiple items in a single variable.

Lists are one of 4 built-in data types in Python used to store collections of data, the other 3 are Tuple, Set, and Dictionary, all with different qualities and usage.

Lists are created using square brackets:

Example

Create a List:

```
thislist = ["apple", "banana", "cherry"]  
print(thislist)
```

List Items

List items are ordered, changeable, and allow duplicate values.

List items are indexed, the first item has index `[0]`, the second item has index `[1]` etc.

Ordered

When we say that lists are ordered, it means that the items have a defined order, and that order will not change.

If you add new items to a list, the new items will be placed at the end of the list.

Note: There are some list methods that will change the order, but in general: the order of the items will not change.

Changeable

The list is changeable, meaning that we can change, add, and remove items in a list after it has been created.

Allow Duplicates

Since lists are indexed, lists can have items with the same value:

Example

Lists allow duplicate values:

```
thislist = ["apple", "banana", "cherry", "apple", "cherry"]  
print(thislist)
```

List Length

To determine how many items a list has, use the `len()` function:

Example

Print the number of items in the list:

```
thislist = ["apple", "banana", "cherry"]  
print(len(thislist))
```

List Items - Data Types

List items can be of any data type:

Example

String, int and boolean data types:

```
list1 = ["apple", "banana", "cherry"]  
list2 = [1, 5, 7, 9, 3]  
list3 = [True, False, False]
```

A list can contain different data types:

Example

A list with strings, integers and boolean values:

```
list1 = ["abc", 34, True, 40, "male"]
```

Python Tuples

Tuples are used to store multiple items in a single variable.

Tuple is one of 4 built-in data types in Python used to store collections of data, the other 3 are List, Set, and Dictionary, all with different qualities and usage.

A tuple is a collection which is ordered and **unchangeable**.

Tuples are written with round brackets.

Example

Create a Tuple:

```
thistuple = ("apple", "banana", "cherry")  
print(thistuple)
```

Tuple Items

Tuple items are ordered, unchangeable, and allow duplicate values.

Tuple items are indexed, the first item has index `[0]`, the second item has index `[1]` etc.

Ordered

When we say that tuples are ordered, it means that the items have a defined order, and that order will not change.

Unchangeable

Tuples are unchangeable, meaning that we cannot change, add or remove items after the tuple has been created.

Allow Duplicates

Since tuples are indexed, they can have items with the same value:

Example

Tuples allow duplicate values:

```
thistuple = ("apple", "banana", "cherry", "apple", "cherry")
print(thistuple)
```

Tuple Length

To determine how many items a tuple has, use the `len()` function:

Example

Print the number of items in the tuple:

```
thistuple = ("apple", "banana", "cherry")
print(len(thistuple))
```

Create Tuple With One Item

To create a tuple with only one item, you have to add a comma after the item, otherwise Python will not recognize it as a tuple.

Example

One item tuple, remember the comma:

```
thistuple = ("apple",)
print(type(thistuple))
```

```
#NOT a tuple
thistuple = ("apple")
print(type(thistuple))
```

Tuple Items - Data Types

Tuple items can be of any data type:

Example

String, int and boolean data types:

```
tuple1 = ("apple", "banana", "cherry")  
tuple2 = (1, 5, 7, 9, 3)  
tuple3 = (True, False, False)
```

A tuple can contain different data types:

Example

A tuple with strings, integers and boolean values:

```
tuple1 = ("abc", 34, True, 40, "male")
```


Python Sets

Set

Sets are used to store multiple items in a single variable.

Set is one of 4 built-in data types in Python used to store collections of data, the other 3 are List, Tuple, and Dictionary, all with different qualities and usage.

A set is a collection which is *unordered*, *unchangeable**, and *unindexed*.

*** Note:** Set *items* are unchangeable, but you can remove items and add new items.

Sets are written with curly brackets.

Example

Create a Set:

```
thisset = {"apple", "banana", "cherry"}  
print(thisset)
```

Note: Sets are unordered, so you cannot be sure in which order the items will appear.

Set Items

Set items are unordered, unchangeable, and do not allow duplicate values

Unordered

Unordered means that the items in a set do not have a defined order.

Set items can appear in a different order every time you use them, and cannot be referred to by index or key.

Unchangeable

Set items are unchangeable, meaning that we cannot change the items after the set has been created.

Once a set is created, you cannot change its items, but you can remove items and add new items.

Duplicates Not Allowed

Sets cannot have two items with the same value.

Example

Duplicate values will be ignored:

```
thisset = {"apple", "banana", "cherry", "apple"}  
  
print(thisset)
```

Note: The values `True` and `1` are considered the same value in sets, and are treated as duplicates:

Example

`True` and `1` is considered the same value:

```
thisset = {"apple", "banana", "cherry", True, 1, 2}  
  
print(thisset)
```

Note: The values `False` and `0` are considered the same value in sets, and are treated as duplicates:

Example

`False` and `0` is considered the same value:

```
thisset = {"apple", "banana", "cherry", False, True, 0}  
  
print(thisset)
```

Get the Length of a Set

To determine how many items a set has, use the `len()` function.

Example

Get the number of items in a set:

```
thisset = {"apple", "banana", "cherry"}
```

```
print(len(thisset))
```

Set Items - Data Types

Set items can be of any data type:

Example

String, int and boolean data types:

```
set1 = {"apple", "banana", "cherry"}
```

```
set2 = {1, 5, 7, 9, 3}
```

```
set3 = {True, False, False}
```

A set can contain different data types:

Example

A set with strings, integers and boolean values:

```
set1 = {"abc", 34, True, 40, "male"}
```

Python Dictionaries

Dictionaries are used to store data values in key:value pairs.

A dictionary is a collection which is ordered*, changeable and do not allow duplicates.

As of Python version 3.7, dictionaries are *ordered*. In Python 3.6 and earlier, dictionaries are *unordered*.

Dictionaries are written with curly brackets, and have keys and values:

Example

Create and print a dictionary:

```
thisdict = {  
    "brand": "Ford",  
    "model": "Mustang",  
    "year": 1964  
}  
print(thisdict)
```

Dictionary Items

Dictionary items are ordered, changeable, and do not allow duplicates.

Dictionary items are presented in key:value pairs, and can be referred to by using the key name.

Example

Print the "brand" value of the dictionary:

```
thisdict = {  
    "brand": "Ford",  
    "model": "Mustang",  
    "year": 1964  
}  
print(thisdict["brand"])
```

Ordered or Unordered?

As of Python version 3.7, dictionaries are *ordered*. In Python 3.6 and earlier, dictionaries are *unordered*.

When we say that dictionaries are ordered, it means that the items have a defined order, and that order will not change.

Unordered means that the items do not have a defined order, you cannot refer to an item by using an index.

Changeable

Dictionaries are changeable, meaning that we can change, add or remove items after the dictionary has been created.

Duplicates Not Allowed

Dictionaries cannot have two items with the same key:

Example

Duplicate values will overwrite existing values:

```
thisdict = {  
    "brand": "Ford",  
    "model": "Mustang",  
    "year": 1964,  
    "year": 2020  
}  
print(thisdict)
```

Dictionary Length

To determine how many items a dictionary has, use the `len()` function:

Example

Print the number of items in the dictionary:

```
print(len(thisdict))
```

Dictionary Items - Data Types

The values in dictionary items can be of any data type:

Example

String, int, boolean, and list data types:

```
thisdict = {  
    "brand": "Ford",  
    "electric": False,  
    "year": 1964,  
    "colors": ["red", "white", "blue"]  
}
```

There are four collection data types in the Python programming language:

- **List** is a collection which is ordered and changeable. Allows duplicate members.
- **Tuple** is a collection which is ordered and unchangeable. Allows duplicate members.
- **Set** is a collection which is unordered, unchangeable*, and unindexed. No duplicate members.
- **Dictionary** is a collection which is ordered** and changeable. No duplicate members.

Python Conditional Statements

Python's conditional statements allow you to execute code based on certain conditions. The most common conditional statements in Python are if, elif, and else. Here are some examples to illustrate their usage:

Basic if statement

Example 1: Simple if statement

```
x = 10
```

```
if x > 5:
```

```
    print("x is greater than 5")
```

In this example, since x is 10, the condition `x > 5` is True, so the code inside the if block is executed, and "x is greater than 5" is printed.

if and else statement

Example 2: if-else statement

```
y = 3
```

```
if y > 5:
```

```
    print("y is greater than 5")
```

```
else:
```

```
    print("y is not greater than 5")
```

Here, since y is 3, the condition `y > 5` is False, so the code inside the else block is executed, and "y is not greater than 5" is printed.

if, elif, and else statement

Example 3: if-elif-else statement

```
z = 5
```

```
if z > 5:
```

```
    print("z is greater than 5")
```

```
elif z == 5:
```

```
    print("z is equal to 5")
```

```
else:
```

```
    print("z is less than 5")
```

In this example, since z is 5, the first condition `z > 5` is False, but the second condition `z == 5` is True, so the code inside the `elif` block is executed, and "z is equal to 5" is printed.

Nested if statements

Example 4: Nested if statements

```
a = 8
```

```
b = 6
```

```
if a > 5:
```

```
    print("a is greater than 5")
```

```
    if b > 5:
```

```
        print("b is also greater than 5")
```

```
    else:
```

```
        print("b is not greater than 5")
```

```
else:
```

```
    print("a is not greater than 5")
```

Here, the condition `a > 5` is True, so the code inside the first if block is executed, printing "a is greater than 5". Then, the nested if checks whether `b > 5`. Since b is 6, it prints "b is also greater than 5".

Combining multiple conditions with logical operators

Example 6: Combining conditions with logical operators

```
d = 7
```

```
e = 10
```

```
if d > 5 and e > 5:
```

```
    print("Both d and e are greater than 5")
```

```
if d > 5 or e > 5:
```

```
    print("At least one of d or e is greater than 5")
```

```
if not d < 5:
```

```
    print("d is not less than 5")
```

Short Hand If

If you have only one statement to execute, you can put it on the same line as the if statement.

Example

One line if statement:

```
if a > b: print("a is greater than b")
```

Short Hand If ... Else

If you have only one statement to execute, one for if, and one for else, you can put it all on the same line:

Example

One line if else statement:

```
a = 2  
b = 330  
print("A") if a > b else print("B")
```

The pass Statement

`if` statements cannot be empty, but if you for some reason have an `if` statement with no content, put in the `pass` statement to avoid getting an error.

Example

```
a = 33  
b = 200  
  
if b > a:  
    pass
```

Python Looping Statements

Python Loops

Python has two primitive loop commands:

- `while` loops
- `for` loops

The while Loop

With the `while` loop we can execute a set of statements as long as a condition is true.

Example

Print i as long as i is less than 6:

```
i = 1
while i < 6:
    print(i)
    i += 1
```

The break Statement

With the `break` statement we can stop the loop even if the while condition is true:

Example

Exit the loop when i is 3:

```
i = 1
while i < 6:
    print(i)
    if i == 3:
        break
    i += 1
```

The continue Statement

With the `continue` statement we can stop the current iteration, and continue with the next:

Example

Continue to the next iteration if i is 3:

```
i = 0
while i < 6:
    i += 1
    if i == 3:
        continue
    print(i)
```

The else Statement

With the `else` statement we can run a block of code once when the condition no longer is true:

Example

Print a message once the condition is false:

```
i = 1
while i < 6:
    print(i)
    i += 1
else:
    print("i is no longer less than 6")
```

Python For Loops

A `for` loop is used for iterating over a sequence (that is either a list, a tuple, a dictionary, a set, or a string).

This is less like the `for` keyword in other programming languages, and works more like an iterator method as found in other object-orientated programming languages.

With the `for` loop we can execute a set of statements, once for each item in a list, tuple, set etc.

The `for` loop does not require an indexing variable to set beforehand. Which means like in any other programming language the indexing variable should be initialized with a value.

Example

Print each fruit in a fruit list:

```
fruits = ["apple", "banana", "cherry"]
for x in fruits:
    print(x)
```

Looping Through a String

Even strings are iterable objects, they contain a sequence of characters:

Example

Loop through the letters in the word "banana":

```
for x in "banana":
    print(x)
```

The range() Function

To loop through a set of code a specified number of times, we can use the `range()` function,

The `range()` function returns a sequence of numbers, starting from 0 by default, and increments by 1 (by default), and ends at a specified number.

Example

Using the range() function:

```
for x in range(6):
    print(x)
```

Note that `range(6)` is not the values of 0 to 6, but the values 0 to 5.

The `range()` function defaults to 0 as a starting value, however it is possible to specify the starting value by adding a parameter: `range(2, 6)`, which means values from 2 to 6 (but not including 6):

Example

Using the start parameter:

```
for x in range(2, 6):
    print(x)
```

The `range()` function defaults to increment the sequence by 1, however it is possible to specify the increment value by adding a third parameter: `range(2, 30, 3)`:

Example

Increment the sequence with 3 (default is 1):

```
for x in range(2, 30, 3):  
    print(x)
```

Else in For Loop

The `else` keyword in a `for` loop specifies a block of code to be executed when the loop is finished:

Example

Print all numbers from 0 to 5, and print a message when the loop has ended:

```
for x in range(6):  
    print(x)  
else:  
    print("Finally finished!")
```

Note: The `else` block will NOT be executed if the loop is stopped by a `break` statement.

Example

Break the loop when `x` is 3, and see what happens with the `else` block:

```
for x in range(6):  
    if x == 3: break  
    print(x)  
else:  
    print("Finally finished!")
```

Nested Loops

A nested loop is a loop inside a loop.

The "inner loop" will be executed one time for each iteration of the "outer loop":

Example

Print each adjective for every fruit:

```
adj = ["red", "big", "tasty"]
fruits = ["apple", "banana", "cherry"]

for x in adj:
    for y in fruits:
        print(x, y)
```